



NETWORKING KEY TERMINOLOGY

FOUNDATION OF MODERN COMPUTER NETWORKS



THEORY

Networking is the backbone of modern communication systems. Different protocols and technologies work together to enable secure communication, file sharing, internet browsing, email transmission, remote access, and network management.

These networking terminologies form the basic foundation required for **learning**:

- ✓ Cyber Security
- ✓ Ethical Hacking
- ✓ Network Administration
- ✓ Cloud Computing
- ✓ System Engineering



Understanding these concepts helps students and professionals identify how devices **communicate**, **exchange data**, and remain **secure** across local and global networks.



KEY PROTOCOL CATEGORIES

COMMUNICATION PROTOCOLS



- HTTP
- FTP
- SMTP
- SSH

SECURITY PROTOCOLS



- WEP
- WPA
- VPN
- IPsec

ROUTING & NETWORK MANAGEMENT



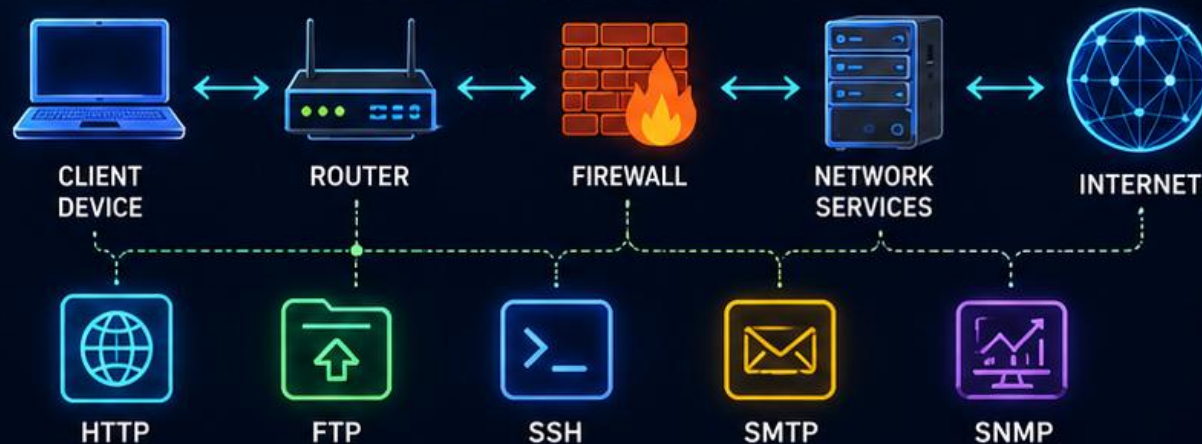
- RIP
- OSPF
- SNMP
- VLAN

REMOTE ACCESS & TUNNELING



- PPTP
- GRE
- RSH

HOW NETWORK COMMUNICATION WORKS





NETWORKING KEY TERMINOLOGY

— Essential Protocols & Their Meanings —



THEORY

Networking protocols are standardized rules that define how data is transmitted, received, and processed across a network. They enable devices, systems, and applications to communicate efficiently and securely.



KEY POINT

Each protocol has a specific purpose. Together, they ensure smooth, secure, and reliable communication across networks.

WHAT YOU'LL LEARN



Secure Communication

Learn how protocols protect data and maintain privacy.



Data Transfer

Understand how files, emails, and web data are transmitted.



Network Management

Explore protocols used for monitoring and managing network devices.



Routing & Connectivity

Discover how data finds the best path across networks.



Remote Access

See how protocols enable secure remote access and tunneling.

COMMON NETWORKING PROTOCOLS AND THEIR PURPOSES

CATEGORY	PROTOCOL	ACRONYM	DESCRIPTION	USE CASE
 COMMUNICATION PROTOCOLS	Hypertext Transfer Protocol	HTTP	Client-server protocol used to send and receive data.	Web browsing
	Simple Mail Transfer Protocol	SMTP	Protocol used to send and receive emails.	Email transmission
	File Transfer Protocol	FTP	Protocol used to transfer files between systems.	File transfer
	Secure Shell	SSH	Secure protocol used to access remote systems.	Remote login
 SECURITY PROTOCOLS	Wi-Fi Protected Access	WPA	Wireless security protocol using strong encryption.	Wi-Fi security
	Virtual Private Network	VPN	Creates a secure encrypted connection over the internet.	Secure remote access
	Internet Protocol Security	IPsec	Secures IP communication through encryption.	VPN & data security
	Extensible Authentication Protocol	EAP	Framework for user authentication methods.	Network authentication
 ROUTING & NETWORK MANAGEMENT	Routing Information Protocol	RIP	Distance-vector protocol for routing in networks.	Small network routing
	Open Shortest Path First	OSPF	Link-state protocol for efficient routing.	Enterprise routing
	Simple Network Management Protocol	SNMP	Protocol used to monitor and manage devices.	Network monitoring
	Virtual Local Area Network	VLAN	Logical segmentation of a network.	Network segmentation
 REMOTE ACCESS & TUNNELING	Point-to-Point Tunneling Protocol	PPTP	Creates a secure tunnel for remote access.	VPN connections
	Generic Routing Encapsulation	GRE	Encapsulates packets for tunneling.	Tunnel management
	Remote Shell	RSH	Allows remote execution of commands.	Remote administration



Understanding these protocols builds a strong foundation for **Cyber Security**, **Network Administration**, and **Modern IT Careers**.

NETWORKING KEY TERMINOLOGY

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HOW NETWORK PROTOCOLS WORK TOGETHER



THEORY

Networking protocols follow a set of rules that allow devices to communicate and exchange data securely and efficiently. Each protocol has a specific role, and together they ensure seamless communication across networks and the internet.

Key Points:

- ✓ Devices use protocols to send, receive and process data.
- ✓ Protocols operate at different layers of the OSI model.
- ✓ Security protocols protect data from unauthorized access.
- ✓ Routing protocols find the best path for data delivery.
- ✓ Management protocols monitor and control network devices.

TYPES OF PROTOCOLS



COMMUNICATION PROTOCOLS

Enable devices to exchange information.

- HTTP • FTP
- SMTP • SSH



SECURITY PROTOCOLS

Protect data and ensure secure communication.

- WPA • IPsec
- VPN • EAP



ROUTING PROTOCOLS

Determine the best path for data delivery.

- RIP • OSPF
- EIGRP • IGRP



MANAGEMENT PROTOCOLS

Monitor and manage network devices.

- SNMP • SYSLOG
- CDP • NTP

THE ROLE OF PROTOCOLS IN NETWORK COMMUNICATION



SENDER
Creates Data



ENCAPSULATION
Data is wrapped with protocol rules



TRANSMISSION
Data travels across the network



SECURITY
Data is protected during transmission



RECEIVER
Data is received and processed

OPERATES AT DIFFERENT LAYERS (OSI MODEL)

7	Application Layer	User applications, protocols like HTTP, FTP, SMTP
6	Presentation Layer	Data format, encryption, compression
5	Session Layer	Session establishment, management, termination
4	Transport Layer	End-to-end communication, TCP, UDP
3	Network Layer	Logical addressing, routing protocols like RIP, OSPF
2	Data Link Layer	MAC addressing, switching, VLAN, STP
1	Physical Layer	Transmission of raw bits over physical media

EXAMPLES OF PROTOCOLS



HTTP, FTP, SMTP, DNS



SSL/TLS



NetBIOS, RPC



TCP, UDP



IP, ICMP, RIP, OSPF



Ethernet, VLAN, STP



Ethernet, Wi-Fi, Fiber



SUMMARY: Networking protocols work together at different layers, each with a unique purpose, to ensure data is delivered securely, accurately, and efficiently across the network.



CONNECT



SECURE



ROUTE



MANAGE



RELY

03

NETWORKING KEY TERMINOLOGY

THE FOUNDATION OF MODERN COMPUTER NETWORKS

Networking Key Terminology includes the essential protocols, technologies, and concepts that form the building blocks of modern communication systems. Understanding these terms helps us design, secure, manage, and troubleshoot networks efficiently.



KEY TAKEAWAYS

- ✓ Protocols define how data is transmitted, received, and processed.
- ✓ Different protocols serve different purposes at different layers.
- ✓ Security protocols protect networks and data from threats.
- ✓ Routing protocols ensure data reaches the destination through the best path.
- ✓ Management protocols help monitor, configure and maintain network devices.
- ✓ These terminologies are the base for advanced networking and cybersecurity concepts.

PROTOCOL CATEGORIES OVERVIEW

COMMUNICATION PROTOCOLS



- HTTP
- FTP
- SMTP
- SSH
- SIP

SECURITY PROTOCOLS



- WPA/WEP
- VPN
- IPsec
- SSL/TLS
- EAP

ROUTING & NETWORK PROTOCOLS



- RIP
- OSPF
- EIGRP
- IGRP
- BGP

MANAGEMENT PROTOCOLS



- SNMP
- SYSLOG
- NTP
- CDP
- SMS

TUNNELING & ACCESS PROTOCOLS



- PPTP
- GRE
- RSH
- TACACS
- RADIUS

WHY THIS IS IMPORTANT?



ENSURES SECURE COMMUNICATION
Security protocols protect data integrity, confidentiality, and availability.



IMPROVES NETWORK PERFORMANCE
Routing protocols find the best and fastest path for data delivery.



EASY MANAGEMENT & MONITORING
Management protocols help administrators monitor and control network devices.



SUPPORTS REMOTE ACCESS
Tunneling and access protocols enable secure remote connections.



BUILDS STRONG IT CAREERS
A strong foundation in these terms opens doors to multiple IT domains.

REAL WORLD APPLICATIONS



BUSINESSES

Use protocols for daily communication, file sharing, emails, and web access.



DATA CENTERS

Rely on routing and security protocols to handle large amounts of data securely.



CYBERSECURITY

Security protocols protect networks from attacks and unauthorized access.



REMOTE USERS

VPN and tunneling protocols enable secure remote access to networks.



IoT DEVICES

Protocols connect and manage smart devices across local and global networks.

BUILD YOUR KNOWLEDGE JOURNEY

1



LEARN BASICS

Understand fundamental protocols and their uses.



PRACTICE LABS

Work on simulations and real-world labs.



SECURE & CONFIGURE

Learn security protocols and device configuration.



TROUBLESHOOT

Identify issues and analyze network behavior.



SPECIALIZE & GROW

Choose your domain and build your IT career.

QUICK REFERENCE (EXAMPLES)

PROTOCOL	ACRONYM	PURPOSE
Hypertext Transfer Protocol	HTTP	Web browsing
File Transfer Protocol	FTP	File transfer
Simple Mail Transfer Protocol	SMTP	Email transmission
Secure Shell	SSH	Secure remote login
Routing Information Protocol	RIP	Routing in networks
Open Shortest Path First	OSPF	Efficient routing
Virtual Private Network	VPN	Secure network access
Internet Protocol Security	IPsec	Secure IP communication



FINAL THOUGHT

Mastering networking key terminology is the first step toward becoming a skilled network professional. Keep learning, keep exploring, and stay updated with new technologies!



COMMON PROTOCOLS (Part 3 of 6)



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TRANSMISSION CONTROL PROTOCOL (TCP)

TCP is a **connection-oriented** protocol that establishes a virtual connection between two devices before transmitting data using a **Three-Way-Handshake**. This connection is maintained until the data transfer is complete, and the devices can continue to send data back and forth as long as the connection is active.



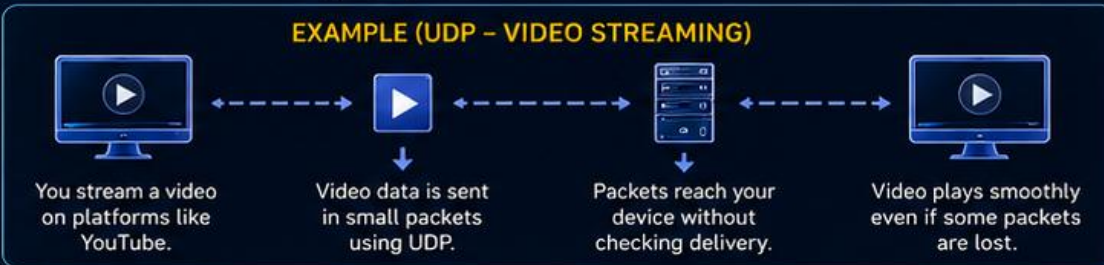
TCP is **reliable** because it ensures delivery and error checking, but it is **slower** than UDP due to connection establishment and additional overhead.

EXAMPLE (TCP - HTTP)



USER DATAGRAM PROTOCOL (UDP)

UDP is a **connectionless** protocol, which means it does not establish a virtual connection before transmitting data. Instead, it sends data packets to the destination without checking if they were received.



UDP is **faster** and more efficient because it has no handshake or acknowledgment process, making it ideal for real-time applications like streaming, VoIP, gaming, and video conferencing.

COMMONLY USED PROTOCOLS OVERVIEW (PART 1)

PROTOCOL	ACRONYM	PORT	DESCRIPTION
Telnet	Telnet	23	Remote login service
Secure Shell	SSH	22	Secure remote login service
Simple Network Management Protocol	SNMP	161-162	Manage network devices
Hyper Text Transfer Protocol	HTTP	80	Used to transfer webpages
Hyper Text Transfer Protocol Secure	HTTPS	443	Used to transfer secure webpages
Domain Name System	DNS	53	Lookup domain names
File Transfer Protocol	FTP	20-21	Used to transfer files
Trivial File Transfer Protocol	TFTP	69	Used to transfer files
Network Time Protocol	NTP	123	Synchronize computer clocks
Simple Mail Transfer Protocol	SMTP	25	Used for email transfer
Post Office Protocol	POP3	110	Used to retrieve emails
Internet Message Access Protocol	IMAP	143	Used to access emails
Server Message Block	SMB	443	Used to transfer files
Network File System	NFS	111, 2049	Used to mount remote systems
Bootstrap Protocol	BOOTP	67, 68	Used to bootstrap computers
Kerberos	Kerberos	88	Used for authentication and authorization
Lightweight Directory Access Protocol	LDAP	389	Used for directory services
Remote Authentication Dial-In User Service	RADIUS	1812, 1813	Used for authentication and authorization
Dynamic Host Configuration Protocol	DHCP	67, 68	Used to configure IP addresses
Remote Desktop Protocol	RDP	3389	Used for remote desktop access
Network News Transfer Protocol	NNTP	119	Used to access newsgroups

PROTOCOL	ACRONYM	PORT	DESCRIPTION
Remote Procedure Call	RPC	135, 137-139	Used to call remote procedures
Identification Protocol	Ident	113	Used to identify user processes
Internet Control Message Protocol	ICMP	0-255	Used to troubleshoot network issues
Internet Group Management Protocol	IGMP	0-255	Used for multicasting
Oracle DB (Default/Alternative) Listener	oracle-tns	1521/1526	Oracle database default/alternative listener
Ingres Lock	ingreslock	1524	Ingres database remote lock service
Squid Web Proxy	http-proxy	3128	Caching and forwarding HTTP web proxy
Secure Copy Protocol	SCP	22	Securely copy files between systems
Session Initiation Protocol	SIP	5060	Used for VoIP sessions
Simple Object Access Protocol	SOAP	80, 443	Used for web services
Secure Socket Layer	SSL	443	Securely transfer files
TCP Wrappers	TCPW	113	Used for access control
Internet Security Association and Key Management Protocol	ISAKMP	500	Used for VPN connections
Microsoft SQL Server	ms-sql-s	1433	Used for client connections to SQL Server
Kerberos Internet Negotiation of Keys	KINK	892	Used for authentication and authorization
Open Shortest Path First	OSPF	89	Used for routing
Point-to-Point Tunneling Protocol	PPTP	1723	Used to create VPNs
Remote Execution	REXEC	512	Execute commands on remote computers
Remote Login	RLOGIN	513	Start interactive shell on remote computer
X Window System	X11	6000	Provides GUI for networked computers
Relational Database Management System	DB2	50000	RDBMS for enterprise applications



COMMON PROTOCOLS



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	Secure Copy Protocol	SCP	22	Securely copy files between systems
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	X Window System	X11	6000	Provides GUI for networked computers
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COMMON PROTOCOLS

(Part 2 of 6)



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#	PROTOCOL	ACRONYM	PORT	DESCRIPTION
21	 Bootstrap Protocol	BOOTP	67, 68	Used to bootstrap computers
22	 Kerberos	Kerberos	88	Used for authentication and authorization
23	 Lightweight Directory Access Protocol	LDAP	389	Used for directory services
24	 Remote Authentication Dial-In User Service	RADIUS	1812, 1813	Used for authentication and authorization
25	 Dynamic Host Configuration Protocol	DHCP	67, 68	Used to configure IP addresses
26	 Remote Desktop Protocol	RDP	3389	Used for remote desktop access
27	 Network News Transfer Protocol	NNTP	119	Used to access newsgroups
28	 Remote Procedure Call	RPC	135, 137-139	Used to call remote procedures
29	 Identification Protocol	Ident	113	Used to identify user processes
30	 Internet Control Message Protocol	ICMP	0-255	Used to troubleshoot network issues
31	 Internet Group Management Protocol	IGMP	0-255	Used for multicasting
32	 Oracle DB (Default/Alternative) Listener	oracle-tns	1521/1526	Oracle database default/alternative listener service
33	 Ingres Lock	ingreslock	1524	Ingres database remote lock service
34	 Squid Web Proxy	http-proxy	3128	Caching and forwarding HTTP web proxy
35	 Secure Copy Protocol	SCP	22	Securely copy files between systems
36	 Session Initiation Protocol	SIP	5060	Used for VoIP sessions
37	 Simple Object Access Protocol	SOAP	80, 443	Used for web services
38	 Secure Socket Layer	SSL	443	Securely transfer files
39	 TCP Wrappers	TCPW	113	Used for access control
40	 Internet Security Association and Key Management Protocol	ISAKMP	500	Used for VPN connections
41	 Microsoft SQL Server	ms-sql-s	1433	Used for client connections to Microsoft SQL Server
42	 Kerberized Internet Negotiation of Keys	KINK	892	Used for authentication and authorization



COMMON PROTOCOLS (Part 3 of 6)



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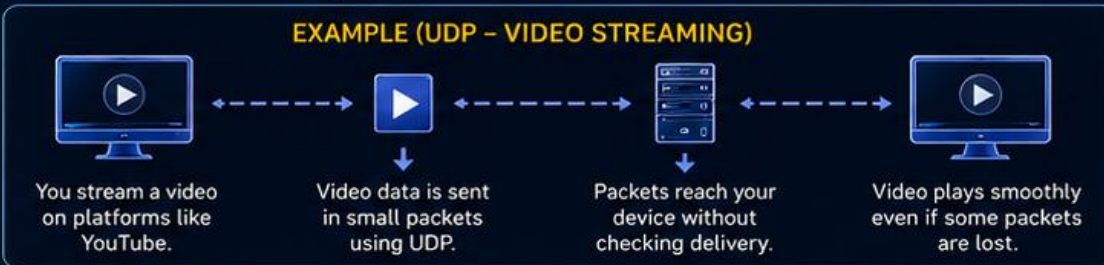
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COMMON PROTOCOLS

(Part 4 of 6)



#	PROTOCOL	ACRONYM	PORT	DESCRIPTION	EXPLANATION
1	 Telnet	Telnet	23	Remote login service	Allows users to remotely access and control a device over a network.
2	 Secure Shell	SSH	22	Secure remote login service	Provides secure encrypted access to a remote device and secure file transfer.
3	 Simple Network Management Protocol	SNMP	161-162	Manage network devices	Used by network managers to monitor and manage network devices.
4	 Hyper Text Transfer Protocol	HTTP	80	Used to transfer webpages	Transfers web pages from a web server to a client (browser).
5	 Hyper Text Transfer Protocol Secure	HTTPS	443	Used to transfer secure webpages	Secure version of HTTP; encrypts data for safe web communication.
6	 Domain Name System	DNS	53	Lookup domain names	Translates domain names (e.g., example.com) into IP addresses.
7	 File Transfer Protocol	FTP	20-21	Used to transfer files	Transfers files between a client and a server over a network.
8	 Trivial File Transfer Protocol	TFTP	69	Used to transfer files	Simpler version of FTP; used for quick file transfers without authentication.
9	 Network Time Protocol	NTP	123	Synchronize computer clocks	Synchronizes the time of devices on a network.
10	 Simple Mail Transfer Protocol	SMTP	25	Used for email transfer	Sends email messages from a client to a mail server or between mail servers.
11	 Post Office Protocol	POP3	110	Used to retrieve emails	Retrieves emails from a mail server and downloads to the client.
12	 Internet Message Access Protocol	IMAP	143	Used to access emails	Accesses emails on the server while keeping them stored online.
13	 Server Message Block	SMB	445	Used to transfer files	Used for file sharing, printer sharing, and other network resource sharing.
14	 Network File System	NFS	111, 2049	Used to mount remote systems	Allows users to access files over a network as if they were local.



COMMON PROTOCOLS

(Part 6 of 6)



#	PROTOCOL	ACRONYM	PORT	DESCRIPTION	EXPLANATION
29	 Secure Copy Protocol	SCP	22	Securely copy files between systems	Uses SSH to securely transfer files between a client and a server.
30	 Session Initiation Protocol	SIP	5060	Used for VoIP sessions	Establishes, modifies, and terminates real-time sessions (voice, video).
31	 Simple Object Access Protocol	SOAP	80, 443	Used for web services	Exchanges structured information in web services using XML.
32	 Secure Socket Layer	SSL	443	Securely transfer files	Provides encryption and security for data transmitted over networks.
33	 TCP Wrappers	TCPW	113	Used for access control	Controls which hosts can connect to services on a system.
34	 Internet Security Association and Key Management Protocol	ISAKMP	500	Used for VPN connections	Establishes and manages IPsec security associations for VPNs.
35	 Microsoft SQL Server	ms-sql-s	1433	Used for client connections to SQL Server	Default port for connecting to Microsoft SQL Server instances.
36	 Kerberized Internet Negotiation of Keys	KINK	892	Used for authentication and authorization	Kerberos protocol extension for secure key negotiation.
37	 Open Shortest Path First	OSPF	89	Used for routing	Link-state routing protocol used within an autonomous system.
38	 Point-to-Point Tunneling Protocol	PPTP	1723	Is used to create VPNs	Creates encrypted tunnels for secure remote access VPNs.
39	 Remote Execution	REXEC	512	Executes commands on remote computers	Runs commands on a remote system and returns the output.
40	 Remote Login	RLOGIN	513	Starts interactive shell on remote computer	Allows users to establish a remote login session.
41	 X Window System	X11	6000	Provides GUI for networked computers	Displays graphical interfaces of remote systems locally.
42	 Relational Database Management System	DB2	50000	RDBMS for enterprise applications	Used for storing, retrieving and managing enterprise data.